

EXAM – SOLUTION

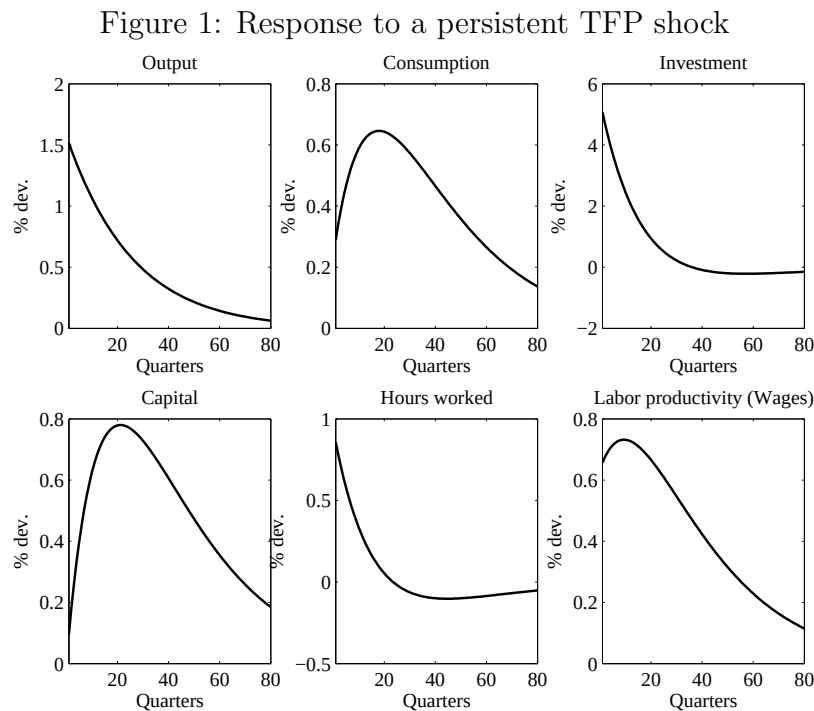
*This is a 3 hours exam. Class slides and any handwritten material are allowed.*  
YOU MUST WRITE LEGIBLY.

I – TRUE, FALSE, UNCERTAIN? (*1/2 of the points*)

For each of those statements, say whether it is true, false, or uncertain and explain why. You should target around 10 to 15 lines per question. Copying the slides will not bring any points.

You should target around 15-20 lines per question. Copying the slides will not bring any points.

1 – Figure 1 below display the responses of some macroeconomic variables to a persistent but not permanent Total Factor Productivity shock in the simple RBC model of the course. Give the economic mechanisms behind those responses. You may want to use the equations that define a competitive equilibrium in that model.



(a) There are two main economic mechanisms that are driving this dynamics: (i) Consumption smoothing : preferences are convex so that agents prefer to smooth consumption over time. Therefore, an increase in today supply of good will not be all consumed today, but partly saved. (ii) Consumption/leisure substitution: consumption and leisure are two normal goods, whose demand increases when income increase or price decrease. The relative price of leisure is the real wage. Therefore, changes in the real wage will affect labor supply through an income and substitution effect.

(b) Given those economic motives, there are two sources of dynamics in the model: (i) The dynamics of the shock: TFP is assumed to be persistent. A shock today has long lasting effects, although non permanent. (ii) The law of motion of capital: capital can be carried from one period to another, and

is the way to transform current goods into future goods.

This is well seen in the case of the analytic RBC model of the slides, whose solution is

$$\begin{aligned}y_{t+1} &= \alpha y_t + z_{t+1} + \alpha \log \alpha \beta \\z_{t+1} &= z_t + \varepsilon_{t+1}\end{aligned}$$

where  $z$  is the exogenous TFP. Even if  $z$  was *iid*, there would be some persistence on the shocks.

(c) The story of those IRF is therefore: TFP increases, but not permanently. This creates a wealth effect as existing capital and labor are more productive *ceteris paribus*. As consumption and output are normal goods, we should expect both to increase. This does not happen to leisure because at the same time leisure becomes more expensive (because marginal productivity of labor increases). This second effect dominates and labor goes up. Not all this extra production ( $y$  is increasing) is going into consumption, and part is saved. hence,  $i$  increases and so does  $k$ . Then all converges back to the steady state.

(d) Finally, note how prices can decentralize those allocations: the real wage is going up (with Cobb-Douglas technology and perfect competition, the real wage moves proportionally to average productivity of labor). The real interest rate (not shown on this graph) is also going up in the short run, so that agents save part of their extra income.

2 – In the MORTENSEN-PISSARIDES' model, the stock of vacancies is a state variable. True? False? Uncertain?

False

(a) In the dynamical systems that we have studied, endogenous state variables are predetermined (in discrete time) or cannot jump (in continuous time). They can only be changed by actions on the control variables.

(b) In MORTENSEN-PISSARIDES, the stock of vacancies is not a state variable. Actually, it is a crucial assumption that helps slowing the model. There is a large number of potential entrepreneurs waiting to post vacancies. There is free entry so that if the value of a vacancy is positive, there will be a discrete entry of new entrepreneurs (the stock of vacancies will jump) so that the value of a vacancy is always zero. The stock of vacancy is therefore determined by a forward looking behavior (pay a posting cost to get some future benefits) + free entry.

3 – Shocks to the capital stock cannot be the main source of fluctuations in a Real Business Cycle model. True? False? Uncertain?

(a) What is the effect of a negative shock to the capital stock? (i) Negative wealth effect:  $c \searrow$ ,  $h \nearrow$ . This negative wealth effect generally dominates the substitution one (leisure is cheaper) so that in general hours  $h$  increase. Overall, we have nevertheless in general  $y \searrow$ . (ii) Consumption smoothing : With less capital, marginal productivity of capital is high, so that households want to invest more. A feasible path would be to consume much less today, invest a lot more so that capital would be rebuilt in one period and consumption would be back to normal tomorrow. That is not optimal as households like to smooth consumption. Therefore, investment will increase smoothly.

(b) In that dynamics,  $c$  and  $y$  on the one side and  $i$  and  $h$  on the other side are negatively correlated (for the “reasonable” calibrations). This is at odds with the pattern of correlations in the business cycle, where all these variables comove positively. Shocks to the capital stock are therefore unlikely to be a story of the business cycle.

(c) It is hard to think of recurrent destruction of the capital stock at business cycle frequencies in the real life (natural disasters are not at the BC frequency, factory fires are not aggregate shocks)

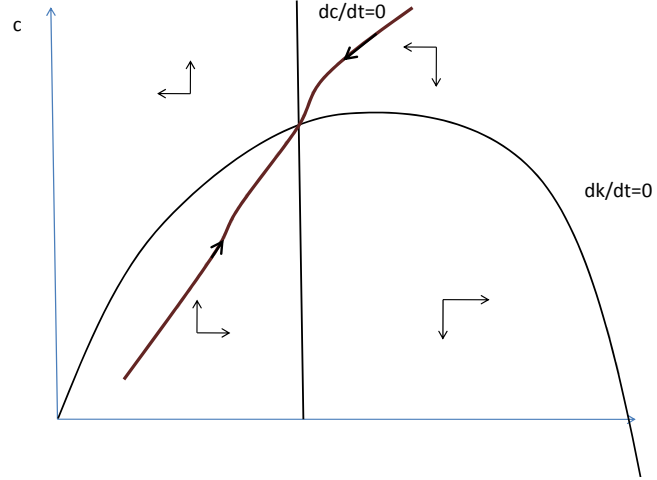
4 – What is the role of the transversality condition in the RAMSEY model?

(a) Mathematically, the RAMSEY model solution is a pair of differential equations in  $(k, c)$ . With

the notations of the slides,

$$\begin{aligned}\dot{k}_t &= \bar{L}f\left(\frac{k_t}{\bar{L}}\right) - (\delta + g)k_t - c_t \\ \frac{\dot{c}_t}{c_t} &= \sigma(f'(\frac{k_t}{\bar{L}}) - \delta - \rho) - g.\end{aligned}$$

Figure 2: Phase diagram of the RAMSEY model



From initial conditions  $(k_0, c_0)$ , the pair of differential equations uniquely determines one trajectory (that we can see using the phase diagram). But if  $k = 0$  is given  $c_0$  is not as it can jump at time 0. The transversality condition  $\lim_{t \rightarrow +\infty} e^{-\rho t} \mu_t K_t = 0$  with  $\mu_t = C_t^{\alpha-1}$  is another boundary condition. The two boundary conditions plus the pair of differential equations uniquely define a trajectory.

(b) From an economic point of view, the TC is an optimality condition. It states that the value of the capital stock at plus infinity has to be zero. It is best to understand this as the limit of a model with finite horizon  $T$  as  $T$  goes to infinity. The TC states that one should not leave capital for the period after the end of the world. Why isn't it optimal? Because another feasible path that would increase consumption at the last period and leave no capital would bring more utility.

(c) Graphically, the TC is ruling out trajectories with too low  $c_0$  that would take the economy to a long run path with zero consumption.

Note: The TC  $\lim_{t \rightarrow +\infty} e^{-\rho t} \mu_t K_t = 0$  is an optimality condition while the No-Ponzi condition  $\lim_{t \rightarrow +\infty} e^{-\rho t} \mu_t K_t \geq 0$  is a constraint imposed to the agents, not a optimality condition.

## II – THE HARROD-DOMAR'S MODEL (1/3 of the points)

The model of HARROD (1939) and DOMAR (1949) can be seen as a version of the SOLOW model but with limited substitution between capital and labor. It produces a result of *instability* of balanced growth. This problem solves the HARROD-DOMAR's model.

Time is continuous. Aggregate production function is

$$Y(t) = \min \left( \frac{K(t)}{v}, \frac{L(t)}{\alpha} \right), \quad \nu > 0, \alpha > 0. \quad (1)$$

$Y(t)$ ,  $K(t)$  and  $L(t)$  are, respectively, aggregate output, the capital stock, and employment.

The saving rate is exogenous and assumed to be  $0 < s < 1$ ; labor force grows exogenously at rate  $n_L$ . The other equations of the model are therefore:

$$S(t) = sY(t) \quad (2)$$

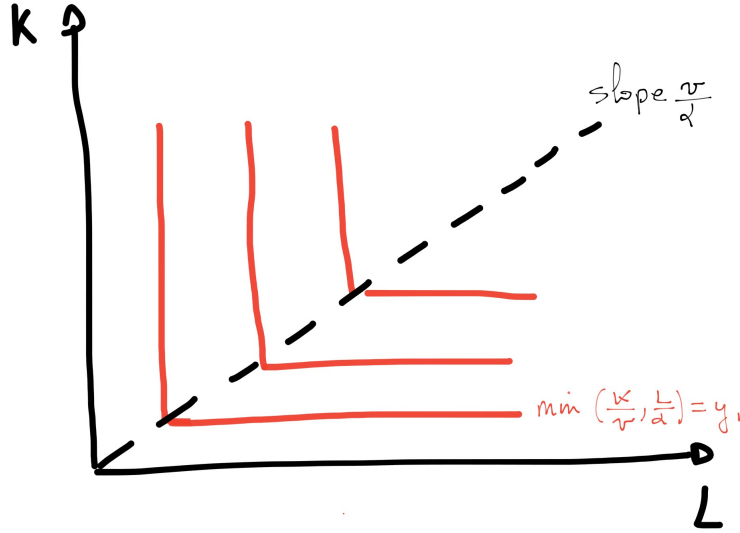
$$I(t) = \delta K(t) + \dot{K}(t) \quad (3)$$

$$I(t) = S(t) \quad (4)$$

$$\frac{\dot{L}(t)}{L(t)} = n_L \quad (5)$$

1 – Draw the isoquants of the production function. Derive expressions for  $Y/K$ ,  $Y/L$ , and  $K/L$  under the assumption that both production factors are fully employed. What happens if the actual  $K/L$  is less than  $v/\alpha$ ? What if it is larger than  $v/\alpha$ ?

Figure 3: Isoquant of the production function in the Harrod-Domar model



There is full employment of both factors when  $\frac{Y}{K} = \frac{1}{v}$  and  $\frac{Y}{L} = \frac{1}{\alpha}$ , which gives  $\frac{K}{L} = \frac{v}{\alpha}$ . In that case, allocations are exactly at the kink of the isoquants.

If  $\frac{K}{L} < \frac{v}{\alpha}$ , then there is unused labor.

If  $\frac{K}{L} > \frac{v}{\alpha}$ , then there is unused capital.

2 – Show that, in order to maintain full employment of capital in the model, output and investment must grow at the so-called “warranted rate of growth” which is equal to  $(s - \delta v)/v$ .

There is full employment of capital when  $Y = \frac{K}{v}$ . This implies  $\dot{Y} = \frac{1}{v}\dot{K}$ . Furthermore,  $I = sY$  implies  $\dot{I} = s\dot{Y}$ .

Now use  $\dot{K} = I - \delta K$ , replace  $\dot{K}$  and  $K$  using the two previous equations to obtain

$$\frac{\dot{Y}}{Y} = \frac{s - \delta v}{sv} \frac{\dot{I}}{I}.$$

$Y$  must grow at rate  $\frac{s - \delta v}{sv}$  (“warranted rate of growth”) to maintain full employment of capital.

3 – Show that, in order to maintain full employment of labour in the model, output and investment must grow at the so-called “natural rate of growth” which is equal to  $n_L$ .

There is full employment of labor if  $Y = \frac{1}{\alpha}L$ . Differencing gives

$$\frac{\dot{Y}}{Y} = \frac{\dot{L}}{L} = n_L.$$

$Y$  must grow at rate  $n_L$  (“natural rate of growth”) to maintain full employment of labor.

4 – Derive the condition under which the economy grows with full employment of both factors of production. This is called the HARROD-DOMAR’s condition. What happens if  $(s - \delta v)/v$  is above or below  $n_L$ ?

The condition for full employment of both factors is

$$n_L = \frac{s - \delta v}{sv}.$$

This is a condition of parameters, and no economic forces are making sure that it holds. Therefore, balanced growth is a knife-edge case, meaning that it holds only for that specific configuration of parameters.

If  $n_L > (s - \delta v)/v$ , there is an increasing unemployment of labor,

If  $n_L < (s - \delta v)/v$ , there is an increasing unemployment of capital.

5 – Show that a HARROD-DOMAR’s like condition also appears in the SOLOW’s model. *Hint: The SOLOW model is described by equations (2) to (5) plus (1) with a general neoclassical production function  $F(K, L)$ . Write the model in per capita terms and derive a condition that involves  $K/L$  and  $n_L$  at the steady state.*

In SOLOW, we have  $\dot{k} = sf(k) - (\delta + n_L)k$ , where  $k = K/L$  and  $f(k) = y = Y/L$ . At the steady state,  $\dot{k} = 0$ , so that  $sy = (\delta + n_L)k$ , or equivalently  $s\frac{y}{k} - \delta = n_L$ . Using the notation  $\frac{y}{k} = \frac{1}{v}$ , we obtain

$$n_L = \frac{s - \delta v}{sv}.$$

6 – Explain why the SOLOW’s model does not suffer from the instability (or knife-edge stability) of the HARROD-DOMAR’s model.

The difference between the two models is that in SOLOW,  $v = \frac{k}{y}$  is *not* a parameter, but an endogenous variable. It is indeed decreasing returns of  $f$  that pins down the steady state level of  $k$  that makes sure that the condition holds.

### III – THE DIAMOND’S COCONUT MODEL (1/3 of the points)

Let’s consider the following environment. Time is continuous. There is a continuum of agents of mass one on an island. The island is composed of a forest and a beach. The forest is where coconut trees grow, while the beach is where people meet. Coconuts are the only good in the economy. Each agent has a linear utility  $u(t)$ . Consuming one coconut gives utility  $y > 0$ , while utility is 0 when the agent is not consuming. Agents discount utility at rate  $r$  (which is also the risk free rate as

utility is linear) and the cost of collecting a coconut is  $c(t)$ . Agents' intertemporal expected utility, as evaluated at date  $T$ , is therefore

$$V(T) = E_T \int_T^\infty e^{-rt} (u(t) - c(t)) dt$$

Production is done by climbing at coconut trees, who are of various height. Agents without coconuts walk in the forest and look for trees. They are said to be *unemployed*. When they are unemployed, they randomly bump into coconut trees. The process of finding a tree is Poisson with rate  $a$ . The cost  $c(t)$  of collecting one coconut is *i.i.d.*, with cdf  $G(c)$ , defined over  $[\underline{c}, \bar{c}]$ , with  $\underline{c} > 0$ . One cannot collect or carry more than one coconut.

There is a taboo on this island, according to which one cannot eat self-collected coconuts. Therefore, once a coconut is collected, an agent has to walk on the beach carrying her coconut, and she will randomly meet another agent with a coconut. When walking in search for a trade partner, agents are said to be *employed*. When a meeting happens, the pair of agents that meets exchange their coconuts (one against one) and eat them. We denote by  $e(t)$  (*employment*) the measure of agents that are holding a coconut and looking for a partner. The process of meeting someone else on the beach is Poisson with rate  $b(e)$ . It is assumed that  $b'(e) > 0$ .

The life of an agent can therefore be described as follows: once agents have a coconut, they simply walk on the beach until they meet another agent with a coconut, and trade. Without a coconut, they walk in the forest until they run into the coconut tree, and then they decide whether to collect the coconut given the cost.

We assume that agents strategy is a stationary policy  $p(c, e)$  which determines a probability of collecting a coconut given its cost  $c \in [\underline{c}, \bar{c}]$  as also a function of the current measure of traders looking for partners,  $e \in [0, 1]$ :

$$p : [\underline{c}, \bar{c}] \times [0, 1] \rightarrow [0, 1]$$

A symmetric equilibrium is a policy  $p$  that is the best response to itself, in the sense that when all other agents use  $p$ , it is optimal for each of them to do so as well.

We will assume that in such an environment, the optimal policy is a reservation cost policy, i.e.,  $p(e(t), c) = 1$  for all  $c \leq c^*(t)$  and  $p(e(t), c) = 0$  otherwise (where we write  $c^*(t)$  instead of  $c^*(e(t))$  to simplify notation).

We denote  $V_E(t)$  the value of holding a coconut and looking for a trade partner (being employed) and  $V_U(t)$  the value of looking for a tree (being unemployed).

**1** – Why don't we have to care about the price of a coconut in a match? Why does it simplify the analysis?

There is only one good in the economy (coconuts) and by assumption one cannot carry more than one coconut. Therefore, trades are always one-for-one. There is no room to split the surplus of a match in a different way (no money, no financial claims, ec...). The price of contents is therefore irrelevant. The fact that we do not need to model the way the surplus is shared within a match simplifies the analysis (we don't need a bargaining theory).

**2** – How to justify  $b'(e) > 0$ ? Explain why this assumption is called "thick market externality"? Why "externality"?

$b'(e) > 0$ : It is easier to bump into someone on the beach if a lot of inhabitants are walking on the beach. It is an externality because no one is taking that effect into account when deciding to climb at a tree.

**3** – Write the arbitrage equation that gives the flow value of being employed. *Hint: by arbitrage, the flow value of being employed should be equal to the expected gain in value of meeting a partner plus the time change of  $V_E(t)$ :*

$$\begin{aligned} \text{flow value } rV_E(t) = & \text{probability of a meeting} \times \left( \text{utility of consumption} + \right. \\ & \left. \text{change in value from employment to unemployment} \right) \\ & + \dot{V}_E(t) \end{aligned} \quad (1)$$

The equation writes

$$rV_E(t) = b(e(t)) \times \left( y + V_U(t) - V_E(t) \right) + \dot{V}_E(t)$$

**4** – Explain why the flow value of unemployment is given by

$$rV_U(t) = a \int_{\underline{c}}^{c^*(t)} (V_E(t) - V_U(t) - c) dG(c) + \dot{V}_U(t) \quad (2)$$

The probability of moving from unemployment to employment is the probability of finding a tree  $a$  times the probability of deciding to climb to collect the coconut. This happens only if  $c$  is lower than  $c^*(t)$ . In that case, the gain in value is  $V_E(t) - V_U(t) - c$ .

**5** – Express the threshold  $c^*(t)$  as a function of  $V_E(t)$  and  $V_U(t)$ .

$c^*(t)$  is such that an individual is indifferent between collecting the coconut and becoming employed (value =  $V_E(t) - c^*(t)$ ) and staying unemployed (value =  $V_U(t)$ ). Hence, we have

$$c^*(t) = V_E(t) - V_U(t).$$

**6** – Given the reservation cost policy, write the law of motion of employment  $e$ , which is here the equivalent of the BEVERIDGE curve in MORTENSEN-PISSARIDES:

$$\dot{e}(t) = \underbrace{a(1 - e(t))G(c^*(t))}_{\text{flow from unemp. to emp.}} - \underbrace{b(e(t))e(t)}_{\text{flow from emp. to unemp.}} \quad (3)$$

**7** – For the rest of the problem, we restrict to steady states of this economy, denoted  $(e^*, c^*)$ . From equations (1) and (2) and the answer to question 5, derive a first implicit relation (A) between  $c^*$  and  $e^*$ . Show that (A) defines a positively sloped locus in the plane  $(e^*, c^*)$  when  $y > c^*$ . Explain why this condition  $y > c^*$  must hold. Admit that this locus  $c = c_A(e)$  is a concave function.

Given that  $c^* = V_E - V_U$ , (1) writes

$$V_E = \frac{b(e)(y - c^*)}{r}.$$

Plug it back in (1) to get

$$V_U = -c^* + b(e) \frac{y - c^*}{r}.$$

(2) writes

$$rV_U = a \int_{\underline{c}}^{c^*} (c^* - c) dG(c),$$

which is equivalent to

$$rV_U = ac^*G(c^*) - a \int_{\underline{c}}^{c^*} cdG(c).$$

Put everything together to get

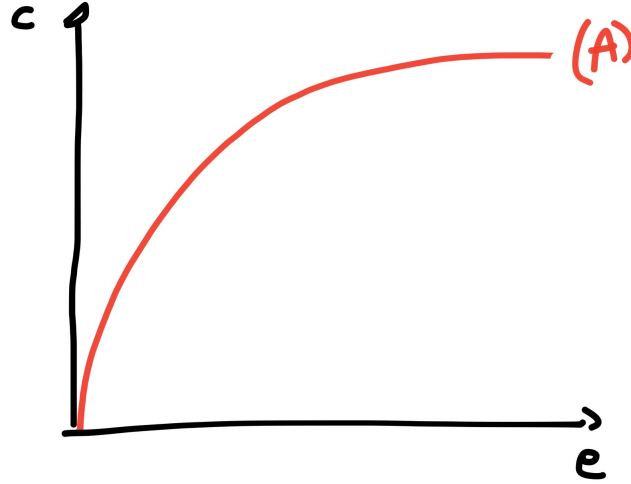
$$c^* = \frac{b(e^*)y + a \int_{\underline{c}}^{c^*} cdG(c)}{r + b(e^*) + aG(c^*)}. \quad (A)$$

Fully differentiate (A) to get

$$(r + b(e^*) + aG(c^*))dc^* = b'(e^*)(y - c^*)de^*.$$

The slope  $dc^*/de^*$  is positive if  $y - c^* > 0$ . It has to be the case as if the threshold cost of grabbing a coconut was higher than the gain from eating it instantaneously, no one would collect it and wait to meet someone to trade. It is admitted that this locus is concave.

Figure 4: The (A) locus



8 – From (3), derive a second implicit relation (B) between  $c^*$  and  $e^*$ . Show that (B) defines a positively sloped locus in the plane  $(e^*, c^*)$ . Denote  $c = c_B(e)$

At the steady states, (3) writes

$$e^* = \frac{aG(c^*)}{aG(c^*) + b(e^*)}.$$

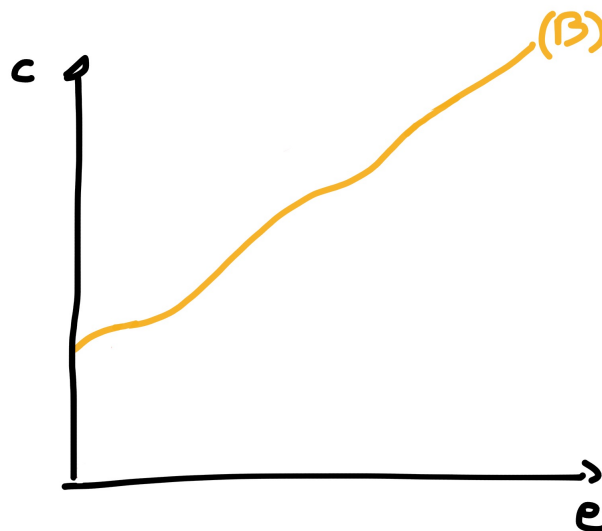
Fully differentiate (B) to get

$$(aG(c^*) + b(e^*) + e^*b'(e^*))de^* = ((1 - e^*)aG'(c^*))dc^*.$$

The slope is positive.



Figure 5: The  $(B)$  locus

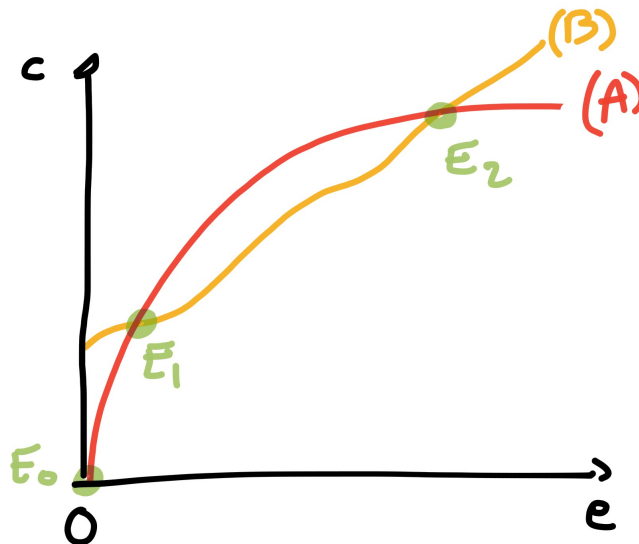


9 – A steady state is an intersection of  $c_A(e)$  and  $c_B(e)$ . Explain why  $(0, \bar{c})$  is always a steady state.

When  $e = 0$ , and with the assumption that  $b(0) = 0$ , there is no chance to meet a trading partner, so that no one will collect a coconut, even when the cost is the minimum one.

10 – Draw  $(A)$  and  $(B)$  in a case of two interior steady states. Explain in words the reason of multiplicity.

Figure 6: Multiple steady states



See Figure 6.

There can be many interior steady states for the following reason. If  $e$  is large, it is optimal to climb high trees to collect coconuts, and  $e$  is indeed large. If  $e$  is small, one would have to wait a long time

before meeting a trading partner on the beach. Therefore, it is not worth climbing high trees, and  $e$  is indeed small.

**11** – Can you Pareto rank an interior steady state and the zero steady state? Explain what is going on.

Take an interior steady state. For all costs  $\tilde{c} < c^*$ , we have a positive utility from being employed, as  $V_E(\tilde{c}) > V(u) + \tilde{c}$ . Therefore, all employed workers have a positive value. As unemployed bear no costs and have a positive probability of becoming employed, they also have a positive value. As all the agents have a zero value at the zero activity steady state, any interior steady state Pareto dominates the zero activity one.

**12** – Is the level of  $e$  suboptimally low in any interior steady state? *Hint: the algebra is quite tedious. Restricting to steady state welfare comparison, you would need to assume a situation in which everybody commits to use a policy  $\hat{p}$  such that  $\hat{p}(c, e^*) = 1$  for all trees with  $c \leq c^* + \delta$  for  $\delta$  positive and small. Denoting  $V_U(\delta)$  and  $V_E(\delta)$  the values in such an economy, you would need to show that  $V'_U(\delta = 0)$  and  $V'_E(\delta = 0)$  are both positive. Unless you have a lot of time left, just explain what is the answer likely to be.*

Intuition: agents do not internalize the fact that they increase the probability to trade for the others when they decide to grab a coconut. Therefore, for a tree of height  $c^*$ , the private value of collecting a coconut is exactly zero but the social one is positive, as collecting that coconut increases  $b$ .

In more details: imagine that a planner forces all agents to accept using a policy such that all  $c \leq c^* + \delta$  are accepted, where  $\delta > 0$  and small. Let us compare steady states (in full generality, one should include the transition). First, denoting  $g = G$ , assuming that  $g(c^*)$  exist and letting  $e^* = e^*(\delta)$ , equation (B) gives

$$\left. \frac{de^*}{d\delta} \right|_{\delta=0} = \frac{ag(c^*)b(e^*)}{(aG(c^*) + b(e^*))^2}$$

. The value of agents searching for trees in steady state is now:

$$rV_u(\delta) = a \int_{\underline{c}}^{c^*+\delta} (V_E(\delta) - V_U(\delta) - c) dG(c).$$

Totally differentiating this with respect to  $\delta$ , we obtain

$$V'_U(\delta = 0) = \frac{aG(c^*)V'_E(\delta = 0)}{r + aG(c^*)}. \quad (\star)$$

Moreover, in steady state:

$$V_E(\delta) = \frac{b(e^*(\delta))(y + V_U(\delta))}{r + b(e^*(\delta))}.$$

Using  $(\star)$ , we have

$$\begin{aligned} V'_E(\delta = 0) &= \frac{b'(e^*)(y + V_U)}{(r + b(e^*))^2} + V'_U(\delta = 0) \\ &= V'_E(\delta = 0) = \frac{b'(e^*)(y + V_U)}{(r + b(e^*))^2} + \frac{aG(c^*)V'_E(\delta = 0)}{r + aG(c^*)} \\ &= \frac{b'(e^*)(r + aG(c^*))(y + V_U)}{r(r + b(e^*))^2} > 0 \end{aligned}$$

which is strictly positive given the thick market externalities, *i.e.*,  $b'(e) > 0$ . Moreover, this implies from  $(\star)$  that  $V'_U(\delta = 0) > 0$ . Consequently, a small increase in  $c^*$  (*i.e.*, in the reservation cost

of agents looking for coconuts) will result in an increase in the values of both types of agents. Therefore, none of the equilibria are constrained Pareto efficient. Some type of subsidy to "search" would improve the welfare of all agents. This is a consequence of the thick market externalities.